

2501-PYL1003 (Mechanics & Special Theory of Relativity)

Practice Mid-Semester Examination — Paper 2 (Unofficial, Instructor-Style)

Full Marks: 60 Time: 2 hours Set: Practice-2

Instructions

An MCQ can also be an MSQ (select all correct options). Justify every choice — no marks for a bare letter. Show all intermediate steps and reasoning.

State any approximation or rounding explicitly, with justification.

Numerical answers should be rounded to two decimal places unless stated otherwise.

1. The energy density of electromagnetic radiation in a cavity in thermal equilibrium (Stefan–Boltzmann-type law) is postulated to depend on temperature Θ , the speed of light c , and the reduced Planck constant $\hbar$ as

$$u = C \Theta^a c^b \hbar^d$$

where u has dimension of energy per unit volume and C is dimensionless. Using $[k_B \Theta] = \text{energy}$ (i.e., treat Θ as carrying dimension ML^2T^{-2} by absorbing k_B), determine a, b, d by dimensional analysis. **[5]**

2 (MSQ). A quantity Π has dimension $\text{M}^0\text{L}^0\text{T}^0$ (i.e., Π is dimensionless). Which of the following is/are valid dimensionless combinations? a) $\frac{v^2}{gL}$, where v is speed, g acceleration, L length (Froude number) b) $\frac{\rho v L}{\eta}$, where ρ is density, η dynamic viscosity (Reynolds number) c) $\frac{Ft}{mv}$, where F is force, t time, m mass, v speed d) $\frac{GM}{c^2 R^2}$, where G is the gravitational constant, M mass, c speed of light, R length **[4]**

3. A particle starts at the origin with initial velocity $v_0 = 12 \text{ m/sec}$ along a straight line. Its acceleration is given as a function of **position** (not time): $a(x) = -2x \text{ m/sec}^2$ (with x in metres), valid for $x \geq 0$ until the particle momentarily stops. (i) Using $a = v \frac{dv}{dx}$, find the maximum distance $x_{\max}$ the particle travels before stopping. (ii) Find the velocity of the particle as a function of x on its way back toward the origin (state its sign). (iii) Is the motion at $x < x_{\max}$ simple harmonic? Justify briefly using the form of $a(x)$.

[3+2+2 = 7]

4. A shell of mass 3 kg is fired vertically upward and simultaneously experiences a constant horizontal wind force of 6 N (in addition to gravity, $g = 10 \text{ m/sec}^2$), starting from rest at the origin at $t = 0$ under the wind force alone in the horizontal direction (i.e., it is *not* launched with horizontal velocity — the wind pushes it as it rises). It is launched with initial vertical speed 40 m/sec. (i) Find the time of flight (return to $y = 0$), neglecting air drag other than the stated constant wind force. (ii) Find the horizontal displacement when it lands. (iii) Find the ratio of the shell's kinetic energy at landing to its kinetic energy at launch. **[2+2+3 = 7]**

5 (MSQ). A point R has cylindrical polar coordinates $(\rho, \phi, z) = (5, 233.13^\circ, -12)$ (angle measured from $+x$ -axis). Which of the following is/are correct about R 's Cartesian and spherical polar (r, θ, ϕ) representations (two decimal places, $z = r \cos \theta$)? a) $x = -3, y = -4$ b) $x = 3, y = -4$ c) $r = 13, \theta = 157.38^\circ$ d) $r = 13, \theta = 112.62^\circ$ **[5]**

6. A vector field is given as $\vec{B} = 2\rho \hat{\rho} - 3\hat{\phi} + 5\hat{z}$ in **cylindrical** polar coordinates (ρ, ϕ, z) . (i) Derive $\hat{\rho}$ and $\hat{\phi}$ in terms of $\hat{i}, \hat{j}$ from first principles (using $\vec{r} = \rho \cos \phi \hat{i} + \rho \sin \phi \hat{j} + z\hat{k}$ and normalizing $\partial \vec{r} / \partial u$). (ii) Evaluate $\vec{B}$ in Cartesian components at the point $\rho = 4, \phi = 60^\circ, z = -1$. (iii) Compute $\vec{B} \cdot \hat{i}$ at that point directly from your Cartesian expression, and also by projecting $\hat{\rho} \cdot \hat{i}$ and $\hat{\phi} \cdot \hat{i}$ onto the given components — confirm they agree. **[3+3+2 = 8]**

7. A satellite is placed in an elliptical orbit about the Earth such that its speed at apogee is exactly 40% of its speed at perigee. (i) Find the eccentricity ϵ of the orbit. (ii) If the perigee distance is $r_p = 7000 \text{ km}$, find the apogee distance r_a . **[3+2 = 5]**

8. A particle moves under a central attractive force $F(r) = -\frac{k}{L^2 r^2}$ (inverse-square, $k > 0$) and is observed to move on a **circular** orbit of radius R . (i) Using $F(R) = \frac{L^2}{mR^3}$ (from the effective-potential turning-point condition, or directly from centripetal requirement), find the angular momentum L in terms of k, m, R . (ii) A small radial perturbation is applied. Derive the condition (in terms of the force law's power-law index, or directly for this $F \propto r^{-2}$ case) for the circular orbit to be **stable**, and show explicitly that the inverse-square orbit satisfies it. **[3+5 = 8]**

9. A projectile is launched due **east** with speed $v_0 = 300 \text{ m/sec}$ at 45° above the horizontal from a location at latitude $\lambda = 45^\circ \text{ N}$. Neglect air resistance; retain only the leading-order Coriolis correction (evaluate using the uncorrected trajectory), and neglect centrifugal effects. (i) Write the three components of the Coriolis acceleration $-2\vec{\Omega} \times \vec{v}$ in a local frame (east, north, up) at general instant t , in terms of v_x (east), v_y (north), v_z (up), Ω , and λ . (ii) Estimate the **north-south** deviation of the landing point from the point directly along the initial launch line, using the zeroth-order (no-Coriolis) trajectory to evaluate the correction. State the direction (north or south) with reasoning. **[3+6 = 9]**

10. A horizontal turntable rotates at constant angular velocity $\Omega = 2 \text{ rad/sec}$. A small puck of mass 0.5 kg sits on the turntable, connected to the rotation axis by a light spring of natural length 0 and spring constant $k = 3 \text{ N/m}$ (i.e., spring force is always $-kr$ directed toward the axis, where r is the puck's distance from the axis), and the puck can slide radially without friction. (i) In the rotating frame, write the radial equation of motion including the centrifugal term. (ii) Determine whether there is a nonzero equilibrium radius $r_{eq} \neq 0$ at which the puck can sit at rest *relative to the turntable*. If so, find it; if not, explain what happens instead in terms of the sign of the net effective radial "force constant." **[3+5 = 8]**

11. A leaky bucket of initial mass M_0 (including water) is pulled straight up at a **constant speed** v_0 by a rope, starting from rest instantaneously at $t = 0^+$. Water leaks out of the bottom at a constant rate $\mu = -dm/dt$ with **zero velocity relative to the bucket** (i.e., the leaking water leaves with the same velocity as the bucket at the moment it leaks — it does not carry away momentum relative to the bucket). (i) Explain why, in this special case (relative exhaust velocity zero), the variable-mass "thrust" term vanishes, and write the resulting equation of motion for the bucket's speed. (ii) Given that the speed must remain constant at v_0 by design (a motor adjusts the pull), find the required tension $T(t)$ in the rope as a function of time, in terms of M_0 , μ , v_0 , and g . **[3+4 = 7]**

12. Sand falls vertically at a constant rate σ (kg/sec) from a stationary hopper onto an empty conveyor belt moving horizontally at constant speed v_{belt} . The sand instantly reaches the belt's horizontal speed upon landing (i.e., friction brings it up to v_{belt} essentially instantaneously in the model), and stays on the belt (no sand falls off). (i) Find the horizontal force the belt motor must exert on the sand (as a function of time, or state that it is constant) to maintain the belt at constant speed, in terms of σ and v_{belt} . (ii) Find the rate of kinetic energy increase of the sand, and compare it to the mechanical power delivered by the motor's horizontal force. Explain, physically, the discrepancy (if any). **[3+4 = 7]**

13. A block B of mass M hangs from a system of pulleys: one fixed pulley attached to the ceiling, and **two** movable pulleys arranged in series (cable from the fixed pulley wraps around movable pulley 1, whose axle also supports movable pulley 2, with the free end of the cable from pulley 2 pulled with force P downward by a person standing on the floor — i.e., a "block and tackle" with 2 movable pulleys giving a 4 : 1 or higher advantage depending on exact routing). You are told the applied force is $P = 15$ N and the system is in equilibrium ($g = 10$ m/sec²). (i) Draw (describe in words/labelled free-body reasoning) the tension distribution across each cable segment, clearly stating your assumed routing, and derive the resulting mechanical advantage. (ii) Compute M and the force that the ceiling attachment point must withstand. **[4+3 = 7]**

14 (MSQ). Two blocks are connected by a light inextensible string over a frictionless pulley at the top of a frictionless incline of angle 37° ($\sin 37^\circ \approx 0.60$, $\cos 37^\circ \approx 0.80$). Block P (mass 4 kg) is on the incline; block Q (mass 3 kg) hangs vertically off the pulley at the top. Which statement(s) is/are correct ($g = 10$ m/sec²)? a) The system accelerates with block Q descending b) The magnitude of acceleration is 2.29 m/sec² (two d.p.) c) The tension in the string exceeds the weight of block Q d) If the incline were instead frictional with $\mu_k = 0.2$ (opposing block P's motion up the incline), the acceleration would decrease in magnitude **[6]**

15. A particle of mass m moves under a central force $F(r) = -kr$ (an attractive **linear** restoring central force, $k > 0$; this is the isotropic 3D harmonic oscillator problem), with nonzero angular momentum L . (i) Write the effective potential $V_{\text{eff}}(r)$ and find the equilibrium radius r_0 of a circular orbit in terms of k , m , L . (ii) By expanding $V_{\text{eff}}(r)$ to second order about r_0 , find the angular frequency of small **radial** oscillations ω_r about the circular orbit, and compare it to the orbital angular frequency $\omega_\theta = L/(mr_0^2)$. Is the resulting orbit (to this order) closed? Justify using the ratio ω_r/ω_θ . **[4+5 = 9]**